Data Communication Questions

1. Why analog to analog signal transmission is important in data communication? (Marks - 2)

2. Why digital to analog signal transmission is important in data communication? (Marks - 2)

3. What is the significance of Quantization and Sampling in Pulse Modulation? (Marks - 3)

4. Calculate the bandwidth of the following modulation where the data rate is 4000 bps and d = 1:
- 4-QAM (Mark 2)
- QPSK (Mark 2)
- 16-QPM (Mark 2)

5. In a 5 channel data lane each lane is transmitting 4000 Kbps data. The unit of data is 1 byte and an extra bit is added at the end of each frame.
a. What is the frame rate (Marks 1)
b. What is the transmission rate high link data lane. (Mark 1)
c. Calculate the output rate and duration (Marks 1)
d. What is the ratio of actual data versus the data being sent in each frame? (Marks 2)

6. Why do we need spreading? Draw two diagrams each from FHSS and DSSS. (Marks 3)

7. What happens when the synchronization bit is out of order with the DEMUX? What will the receiver get with a mismatched synchronization? (Marks - 2)

8. In Statistical Time Division Multiplexing how do scientists fight the 'empty slot' problem and why reducing the frame size is often better than synchronization time division multiplexing? (Marks 2)